



मालवीय राष्ट्रीय प्रौद्योगिकी संस्थान जयपुर (राष्ट्रीय महत्व का संस्थान)
Malaviya National Institute of Technology Jaipur (An Institute of National
Importance)

End Term Examination 2024-25, B. Tech (Semester-II)

Date: 06-05-2025

Paper Name: Logic System Design
Paper Code: 22CST152

Maximum Marks: 50

Maximum Time: 2 Hrs 30 Minutes

Note:

Information provided in the questions are assumed to be sufficient and **no doubts shall be entertained during the examination**. Suitable and/or proper assumption(s) may be taken into consideration, if applicable.

Answer the following questions. Answers must be precise and specific to the points.

Q1.(a) Compare the functionalities of SR and JK flip-flops. Construct a D latch using SR latch and also show the corresponding state table. [2+3]

(b) Design/Implement the following function using a suitable multiplexer
 $f_{a,b,c,d} = \sum(0,3,5,6,8,9,10,11,12,13,15)$, considering a, c and d as selection lines in the multiplexer, and also justify the functionalities. [3+2]

Q2.(a) Consider two boolean variables x and y. Using the Venn diagram proves the following equalities. $(x+y)' = x'y'$ and $(xy)' = x'y' + y'$ [2+2]

(b) Given half-adder circuit(s), implement a full adder circuit using those half-adder(s). Also, derive the boolean functions S_f and C_f in terms of S_h and C_h . [3+3]

Note: S_{hf} and C_{hf} represent the sum and carry in a half/full adder circuit, respectively.

Q3.(a) Implement the following boolean functions $f_{a,b,c} = \sum(0,6)$ and $f_{a,b,c,d,e} = (a+b)(c+d)e$ using NAND (AND-Invert only) and NOR (OR-Invert only) gates, respectively. [3+2]

(b) Design a 4x16 decoder using 3x8 decoder(s) and draw the gate level implementation of the final circuit. [2+3]

Q4.(a) What is the carry propagation delay in a parallel adder? Propose and demonstrate a mechanism to solve the aforesaid problem. [3+2]

(b) Design a Modulo-4 counter (right shift register) and illustrate the corresponding timing diagram, assuming the initial state $Q_1Q_2Q_3Q_4 = 0001$. [3+2]

Q5.(a) What is a ripple counter? Design a 3 bit binary asynchronous up counter using T flip-flops and also show the sequence by demonstrating the timing diagram [1+2+2]

(b) Given the following function, find out the corresponding truth table. Also deduce the function in POS form starting from considering minterms only.

$$f_{a,b,c,d} = a'd + ac + cb' + bd + ab'd'$$

[3+2]

-----X--X--X-----